

Walk-unbiased doubly robust graph propagation for recommendation from exposure-biased logs

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support code development and the development of ideas for the paper. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

All data sets are public: Coat (<https://www.cs.cornell.edu/~schnabts/mnar/>), Yahoo! R3 (Yahoo! Webscope; the version distributed with the AutoDebias code, <https://github.com/DongHande/AutoDebias>) and KuaiRec 2.0 (<https://zenodo.org/records/18164998>).

Code, configurations and all result files are available at https://github.com/vinhqdang/causal_learning_graph_recommendation.